

Emanuel Nader

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EDUCATION

University of California, San Diego

Expected June 2028

B.S. in Mathematics-Computer Science

La Jolla, CA

- Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Systems Programming, Object-Oriented Design

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, HTML/CSS

Frameworks/Libraries: React, Vite, Tailwind, Node.js, Express.js, Next.js, Flask, FastAPI, OpenAI API

Cloud and Databases: AWS, PostgreSQL, MongoDB, SQLite, Redis, Firebase, Supabase, pgvector, Vercel

DevOps and Tools: Docker, Nginx, GitHub Actions, Prometheus, Grafana, Locust, CI/CD, Git, Linux, Figma

Concepts: REST APIs, Microservices, Kalman Filtering, Unit/Integration Testing, pytest, System Design, Agile/Scrum

EXPERIENCE

Tesla

Jun 2026 – Present

Technical Program Manager Intern – Chassis Controls

Palo Alto, CA

- Rebuilt and shipped an internal fleet-planning console in React/Vite/Tailwind on a Flask REST API and SQLite, consolidating 4 planner workflows into one interface and cutting manual cross-checking by 75% for 7 Vehicle Software teams.
- Designed a weighted release-scoring algorithm ranking 350+ vehicles 0 to 100 on 5 metrics with scarcity-protection caps, turning quarterly decommission review into a ranked list guiding 25-45 releases per quarter and recovering fleet carrying cost.
- Implemented config-driven eligibility validation on hardware-pedigree and firmware-branch predicates, flagging a significant share of the campaign population as unsupported so parts and technician hours went only to reworkable vehicles.
- Reconciled 3 internal systems into 4-stage source-of-truth tables tracking approval, parts, technician, and completion, eliminating ownership ambiguity across 2 concurrent multi-site retrofit campaigns.

HP Tech Ventures

Feb 2026 – Apr 2026

Software Engineering Extern (Technical Analyst)

Remote

- Cut due diligence cycle time by 40%, saving 15+ hours/week, by building Python scrapers that aggregated and cleaned 10,000+ market records into structured investment reports.
- Halved technical risk analysis time (4 hrs to under 2 hrs/report) by building an LLM-powered research tool using NLP pipelines to auto-generate structured competitive landscape briefs.

Theta Tau - Professional Engineering Fraternity

May 2025 – Present

Software Project Lead

San Diego, CA

- Led Agile delivery of Garnett, a production React/Node.js app for 70+ members, maintaining 99%+ uptime and supporting 500+ monthly transactions across dues, events, and merit tracking.
- Drove 100% on-time delivery across 2 full-stack projects (Nutribase, LockedIn) by mentoring 12 engineers through structured code reviews, sprint planning, and technical design sessions.

PROJECTS

trackbench | C++17, Python, PostgreSQL, React, Docker, nuScenes | GitHub

August 2026

- Cut multi-object tracking identity switches by 53% (890 → 415) and improved MOTA from -1.351 to +0.666 by building a C++17 Constant-Velocity EKF pipeline with Hungarian/greedy association solvers on nuScenes driving data.
- Accelerated association solver throughput by 29× (p99: 0.531ms → 0.018ms) by implementing a greedy matching algorithm validated across a 24-cell ablation grid of 3,840 benchmark runs with CI regression gating.

ShortStack | Flask, PostgreSQL, Redis, Nginx, Docker, Prometheus, GitHub Actions

April 2026

- Reduced redirect latency by 93% (30ms to under 2ms on cache hits) by integrating Redis caching across 3 load-balanced Nginx replicas, validated under 500 concurrent users via Locust load testing.
- Achieved 84% test coverage across 25 automated tests by building a full CI pipeline with GitHub Actions and orchestrating 7 Docker containers with health checks, auto-restart policies, and Prometheus/Grafana monitoring.

HealthBook | Next.js, FastAPI, Browser Use SDK, OpenAI Realtime API | GitHub

March 2026

- Reduced medical appointment booking time 90% (30 min to under 3 min) by orchestrating a Browser Use SDK agent to automate Zocdoc navigation and trigger real AI phone calls to doctor offices via OpenAI Realtime API and Twilio.
- Eliminated repeated health history entry across sessions by engineering a pgvector semantic memory pipeline in Supabase, retrieving the top 3 relevant memory chunks via cosine similarity and injecting them into each agent task prompt.

Nutribase | MongoDB, Express.js, React, Node.js, C++ | GitHub

May 2025

- Reduced food waste tracking errors and unauthorized access by 60% by engineering a MERN-stack Smart Fridge Tracker with secure, role-based authentication and user analytics dashboards.
- Programmed custom ESP32 firmware in C++ to stream real-time weight data (10+ payloads/second) to a scalable Node.js backend via low-latency MQTT/HTTP pipelines.